

Mock Olympiad Solutions

PRAKRIT GAJUREL

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As two clowns at the mercy of fate, let's fight this out
fair and square.

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Here's the set of solutions I've compiled for the SOMAES and MIN Physics and Math mock olympiad happening on August 16. These solutions are intended as a reference, and students may use different valid approaches. If something in a solution in this file is unclear, feel free to assume it and move on with the rest of the solution. In grading, however, any mathematically sound solution should receive full credit.

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§1 Official Solutions

§1.1 Solution/s to Problem 1

Problem 1.1. Triangle MIN is isosceles with $MI = MN$. Medians $\overline{IB}$ and $\overline{NA}$ are perpendicular to each other, and $IB = NA = 12$. Find, with proof, the area of $\triangle MIN$.

Remark 1.2. In what follows, we denote by $[X]$ the area contained within a polygon X .

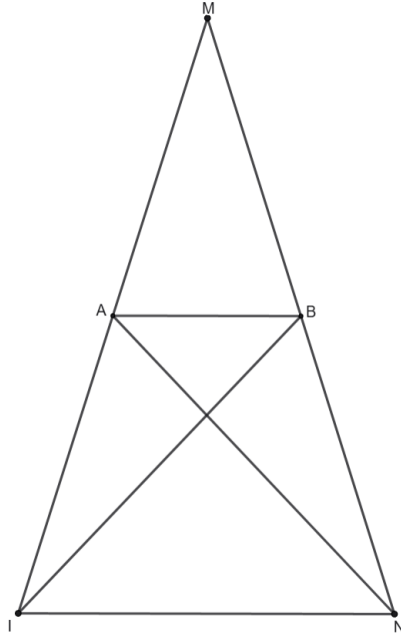


Figure 1: Diagram for the problem.

§1.1.1 Solution 1(areas)

Proof. We show that the answer is 96.

It's easy to observe that $[IABN] = 3[MBA]$, and also,

$$[IABN] = \frac{1}{2}(IB \cdot NA) = 72$$

Giving $[MIN] = \frac{4}{3}[IABN] = 96$. □

§1.1.2 Solution 2(centroid properties)

Proof. Let $G := IB \cap NA$ and let C be the midpoint of IN . As G is the centroid, it follows that $\frac{NG}{GA} = \frac{2}{1}$ giving $NG = 8$ and $GA = 4$, similarly we obtain $IG = 8 = 2GB$. Pythagoras on $\triangle AGI$ gives $IA = AM = BM = NB = 4\sqrt{5}$, and Pythagoras on $\triangle IGN$ gives $IN = 8\sqrt{2} = 2CN$. Again, Pythagoras on $\triangle MNC$ gives $MC = 12\sqrt{2}$, and we get our required answer as:

$$[MIN] = \frac{1}{2}(MC \cdot IN) = \frac{1}{2}(12\sqrt{2} \cdot 8\sqrt{2}) = 96$$

as desired. □

§1.1.3 Solution 3(cartesian bash)

Proof. We proceed in the Cartesian plane. Without loss of generality, let $I = (0, 0)$ and $N = (n, 0)$, giving $M = \left(\frac{n}{2}, h\right)$. Since A and B are the midpoints of MI and MN , we obtain:

$$A = \left(\frac{n}{4}, \frac{h}{2}\right), \quad B = \left(\frac{3n}{4}, \frac{h}{2}\right)$$

Also, we obtain:

$$\overrightarrow{IB} = \left\langle \frac{3n}{4}, \frac{h}{2} \right\rangle, \quad \overrightarrow{NA} = \left\langle -\frac{3n}{4}, \frac{h}{2} \right\rangle$$

As $IB \perp NA$, we get that $h = \frac{3n}{2}$. Also using that $IB = 12$ gives $n^2 = 128$ by the distance formula, giving:

$$[MIN] = \frac{1}{2}(nh) = \frac{1}{2} \left(\frac{3n^2}{2} \right) = \boxed{96}.$$

□

§1.2 Solution/s to Problem 2

Problem 1.3. You are given that x and y are real numbers.

1. Given that $x + y$ and $x + y^2$ are rational numbers, is it always true that x and y are also rational numbers?
2. Given that $x + y$, $x + y^2$, and $x + y^3$ are rational numbers, is it always true that x and y are also rational numbers?

§1.2.1 Solution 1

Proof. Part (1) is pretty straightforward. Obviously $\sqrt{p}$ for any prime p is always irrational, and just considering something like $x = -\frac{\sqrt{p}}{2}$ and $y = \frac{1 + \sqrt{p}}{2}$ gives us that the answer is no.

For part (2), the answer is yes. Define the following:

$$A = (x + y^2) - (x + y) = y(y - 1), \quad B = (x + y^3) - (x + y^2) = y^2(y - 1)$$

Obviously A and B are rational. When $A = 0$, we have $y(y - 1) = 0$ giving $y = 0$ or $y = 1$ giving y rational, and so is $x = (x + y) - y$. When $A \neq 0$, $\frac{B}{A} = y$ is obviously rational, and of course, so is $x = (x + y) - y$. Done. $\square$

§1.3 Solution/s to Problem 3

Problem 1.4. For every positive integer n , express, in terms of n , the number of n -digit positive integers satisfying both of the following conditions:

- No two consecutive digits are equal.
- The last digit is prime.

§1.3.1 Solution 1

Proof. Let a_n denote the number of valid n -digit numbers whose last digit is prime. We claim that the answer is $a_n = \frac{2}{5}(9^n - (-1)^n)$. We first begin by a counting argument.

If we have the n -digit number, the last digit must be one of 2, 3, 5, or 7. Let us consider a single one of these. Then, the second last digit can be any number from 0 to 9 excluding the prime digit we chose, giving us 9 choices. Continuing this for the rest of the $n - 1$ digits gives us an answer of 9^{n-1} , but we had 4 choices for our primes so it is $4 \cdot 9^{n-1}$. We observe, however, that the first digit could end up being 0, giving us NOT the value of a_n but the value of a_{n-1} and so to account for those cases, we add a_{n-1} to get our recursion $a_n + a_{n-1} = 4 \cdot 9^{n-1}$.

Solving the recursion, we get:

$$\begin{aligned}
 a_n &= 4 \cdot 9^{n-1} - a_{n-1} \\
 &= 4 \cdot 9^{n-1} - 4 \cdot 9^{n-2} + a_{n-2} \\
 &= 4(9^{n-1} - 9^{n-2} + 9^{n-3} - \dots) \\
 &= 4 \left(\frac{9^n - (-1)^n}{10} \right) \\
 &= \frac{2}{5}(9^n - (-1)^n)
 \end{aligned}$$

We're done! □

§1.4 Solution/s to Problem 4

Problem 1.5. Ram and Hari have created a program that, given a list of n numbers $a_1, a_2, \dots, a_n$, calculates a list $b_1, b_2, \dots, b_n$ such that, for each $1 \leq k \leq n$, the number b_k is the number of times a_k appears in the list. For example, the list

1, 2, 3, 1

outputs

2, 1, 1, 2.

Next, Ram asks Hari to run the program on a list of 2083 numbers. He then asks him to apply the program to the resulting list, and so on, until a number greater than or equal to k appears in the list. Find, with proof, the largest value of k for which, whatever the initial list of 2083 positive integers is, it is always possible for Hari to do what Ram asked.

§1.4.1 Solution 1

Proof. Define the *frequency* of a number x to be the number of times x appears in a list.

We begin by proving the following claim:

Claim 1.6 — The sequence of the frequencies for each position of the list is bounded and non-decreasing.

Proof. We first show that the sequence is bounded. Since the list contains exactly $n = 2083$ numbers, a number can appear at most 2083 times. Also, a number must appear at least 1 time in the list. Therefore, every number generated must be between 1 and 2083, giving us that our sequence is bounded.

Consider in the first step, any number x that appears c times. Then, in the list generated, the number c will be present at least c times. The "at least" portion comes from the fact that multiple distinct values in the original list could have the same frequency c . Again, consider that number c in the new list that appears d times. So, in the next list, all of the c 's turn into d 's, which guarantees that the frequency of d in the new list is e where $e \geq d$. Therefore, the sequence of the frequencies of the elements is non-decreasing. $\square$

Since that sequence is bounded above and is non-decreasing, it must eventually reach a constant state. Obviously, in that constant list, any number x will appear exactly x times. We claim that the maximum possible value of k is $k = 65$.

For our construction, just consider the following sequence:

1, 2, 2, 3, 3, 3, $\dots$, 65, 65

excluding all 62 of the 62s. This construction obviously works as the total number of items in the list is $\frac{65 \cdot 66}{2} - 62 = 2083$, so we're done for the construction.

We now show that 65 is indeed the maximum. For the sake of contradiction, assume that we have a valid list for when it becomes constant such that all the numbers in the list are strictly less than 65. Then, let the distinct numbers be denoted by $n_1, n_2, \dots, n_m$ where $m \leq 64$ assuming without loss of generality that $n_1 < n_2 < \dots < n_m$. This should give us that $n_1 + n_2 + \dots + n_m = 2083$, however,

$$n_1 < n_2 < \dots < n_m < 65 \implies n_1 + n_2 + \dots + n_m \leq 1 + 2 + \dots + 64 < 2083,$$

Contradiction! Thus, 65 is the required answer. □